

SOFTWARE TESTING

MASTERCLASS

Equivalence Partition and Boundary Value Analysis

Advance Testing Concepts

Equivalence Partition and Boundary Value Analysis

Black box testing is a method of testing software in which the tester is unaware of the internal workings. In these techniques, we test the software from the end users' perspective.

Cover the most part with the least effort



Advance Testing Concepts

Equivalence Partition and Boundary Value Analysis

- Equivalence Partitioning
- Boundary Value Analysis
- Decision Table Testing
- State Transition Testing
- Use case testing



Advance Testing Concepts

Equivalence Partition and Boundary Value Analysis

Equivalence Partitioning is a powerful testing technique that focuses on eliminating sets of input data that result in the same system behavior and produce identical outcomes during application testing.

Reduce
repetitive
test cases





Advance Testing Concepts

Equivalence Partition and Boundary Value Analysis

Requirement: The number field should accept integer values within the range of -5000 to 5000

-5000

0

5000





Advance Testing Concepts

Equivalence Partition and Boundary Value Analysis



TC 1: -100

TC 2: 0

TC 3: 100

Focus on
behaviour
and response
of the system



Advance Testing Concepts

Equivalence Partition and Boundary Value Analysis



Drag and drop video files to upload
Your videos will be private until you publish them.

SELECT FILES

- Testing of an online video upload website that accepts different file formats, such as .mp4, .mov, and .mkv.
- Equivalence Partitioning by grouping the file formats into distinct partitions.
- By selecting representative files from each partition, we can cover the behavior and functionality of the website



Advance Testing Concepts

Equivalence Partition and Boundary Value Analysis

BVA focuses on testing the boundaries of input values, as more issues are often found at the boundaries.

IT complements
Equivalence Partitioning

By testing the edges of the equivalence classes, which include the boundaries, we can uncover hidden defects





Advance Testing Concepts

Equivalence Partition and Boundary Value Analysis

- **2-Value Boundary Value Analysis:** In two-value analysis, the boundary value and the value that is just over the boundary (by the smallest possible increment) are used
- Requirement: Field can accepts values between 1 and 100
- Test Cases: 0, 1 , 100, 101





Advance Testing Concepts

Equivalence Partition and Boundary Value Analysis

- **2-Value Boundary Value Analysis:** the boundary value and the value that is just over the boundary by the smallest possible increment are used
- Requirement: Field can accepts values between 1 and 100
- Test Cases: 0, 1 , 100, 101





Advance Testing Concepts

Equivalence Partition and Boundary Value Analysis

- **3-Value Boundary Value Analysis:** It expands upon the 2-value approach by including an additional test case within each boundary. We test not only invalid values but also valid values just above those boundaries
- Requirement: Field can accept values between 1 and 100
- Test Cases: 0, 1, 2, 99, 100, 101

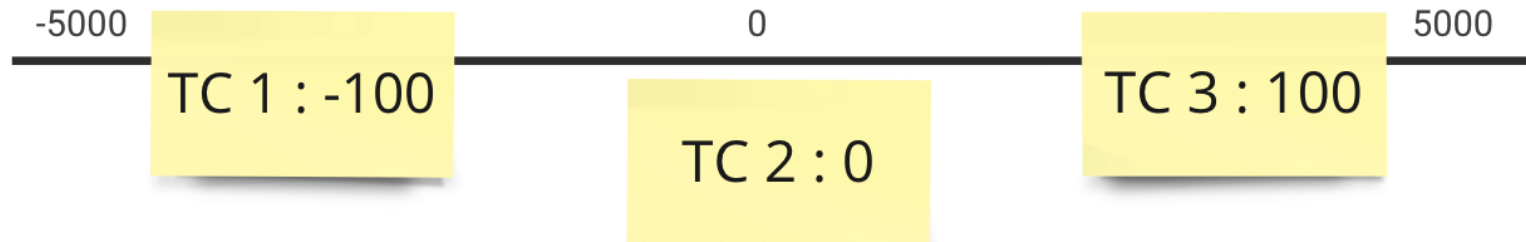




Advance Testing Concepts

Equivalence Partition and Boundary Value Analysis

Requirement: The number field should accept integer values within the range of -5000 to 5000

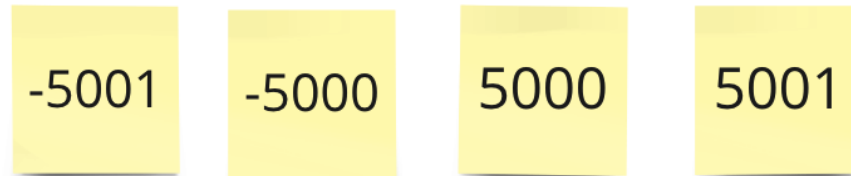


Advance Testing Concepts

Equivalence Partition and Boundary Value Analysis



2-value analysis:



3-value analysis:

